DATA RIDE



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DATA RIDE

Data Ride uses **open-source technology** and **cyclist-generated data** to reveal real cycling conditions and infrastructure gaps in the Lorraine region. The project starts by engaging local stakeholders in Nancy through a co-created mapping, while a senseBox:bike prototype is developed in Münster.

We then run a **data collection campaign** and transform the results into visual material to support a **local advocacy strategy**. Finally, we launch the advocacy campaign together with partners and produce collaboratively a policy recommendations document.

The project concludes with advocacy sustainability guidelines and the **Data Ride Advocacy Toolkit**, designed for broader dissemination and long-term inspiration.

DATA RIDE overview

Nancy's landscape of cycling practices mapping

Contact main actors → 'Snowball'to enrich the mapping

Validate hypotheses that can be dada-tested

Organise who would be up to collect this data, cycling with senseBOX → Installation of the prototype in the bikes and data collection - Community Sensing Campaign with local cyclists

In parallel: Software/Hardware prototype development in Münster and tested in Germany

Publish of the Tech. materials in Open Source license

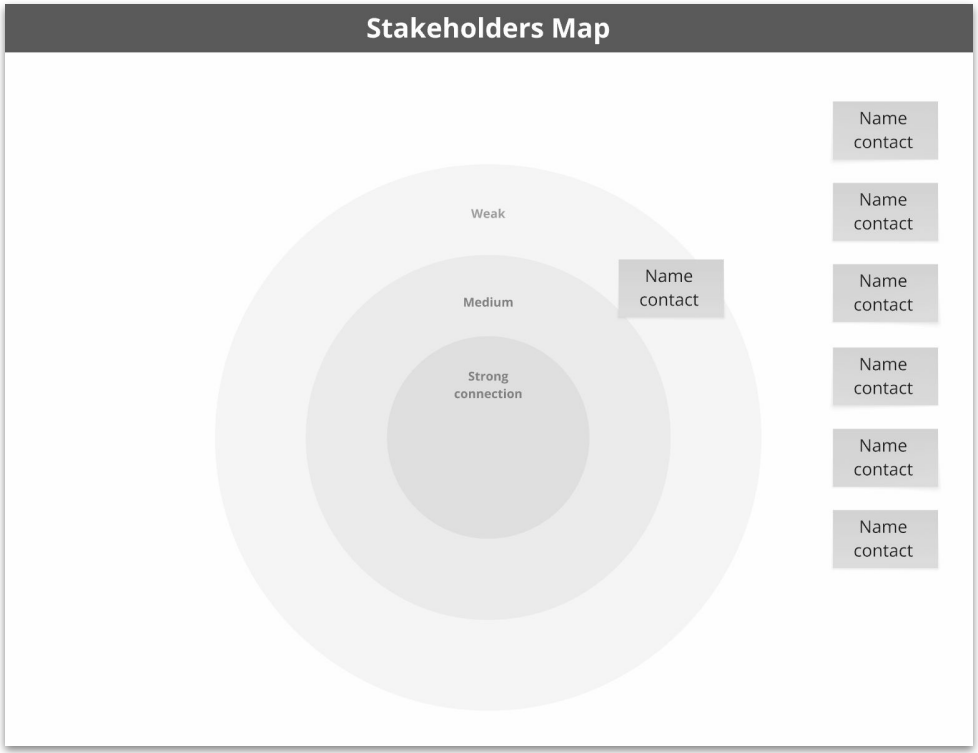
Data analysis and collective preparation of the advocacy campaign with local actors

We are here!

Development of the advocacy campaign strategy, including communication materials such as posters, stickers, online reach, politicians contacts

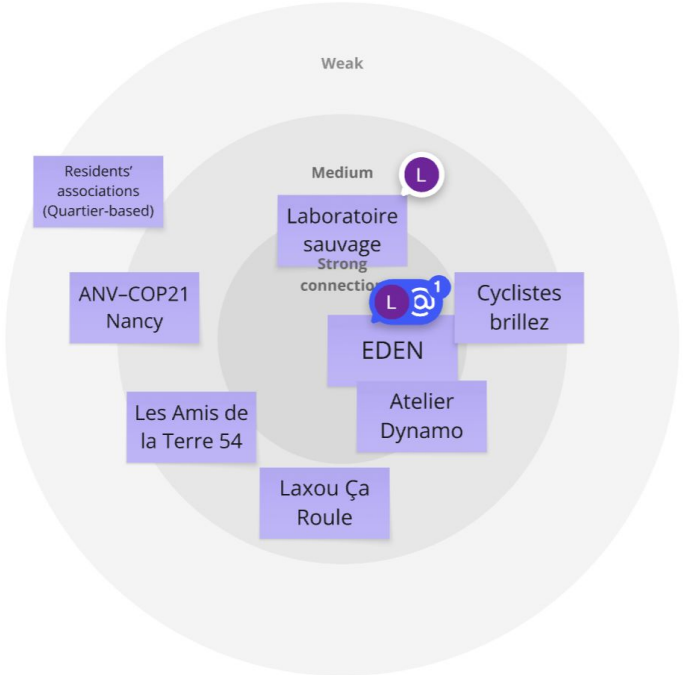
Campaign kick-off and preparation of a policy recommendations' draft → Advocacy sustainability plan with motivated local actors, final policy recommendations document → Data Ride Advocacy toolkit for dissemination and inspiration

Nancy’s landscape of cycling practices



https://miro.com/app/board/uXjVJuWNhVw=/

Citizen-led groups (Neighbourhood groups, associations, NGOs, etc)



Name contact

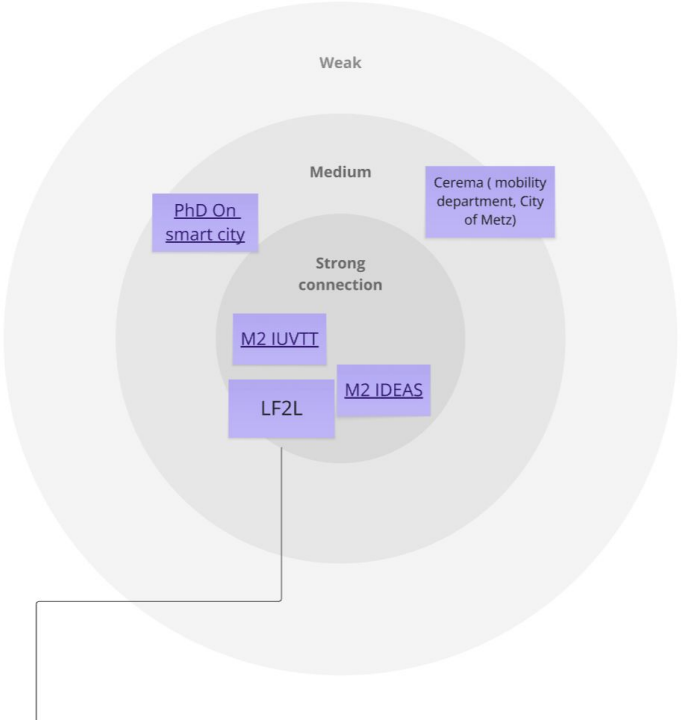
Name contact

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Name contact

Name contact

Institutions (incl. Universities, Schools etc)



Name contact

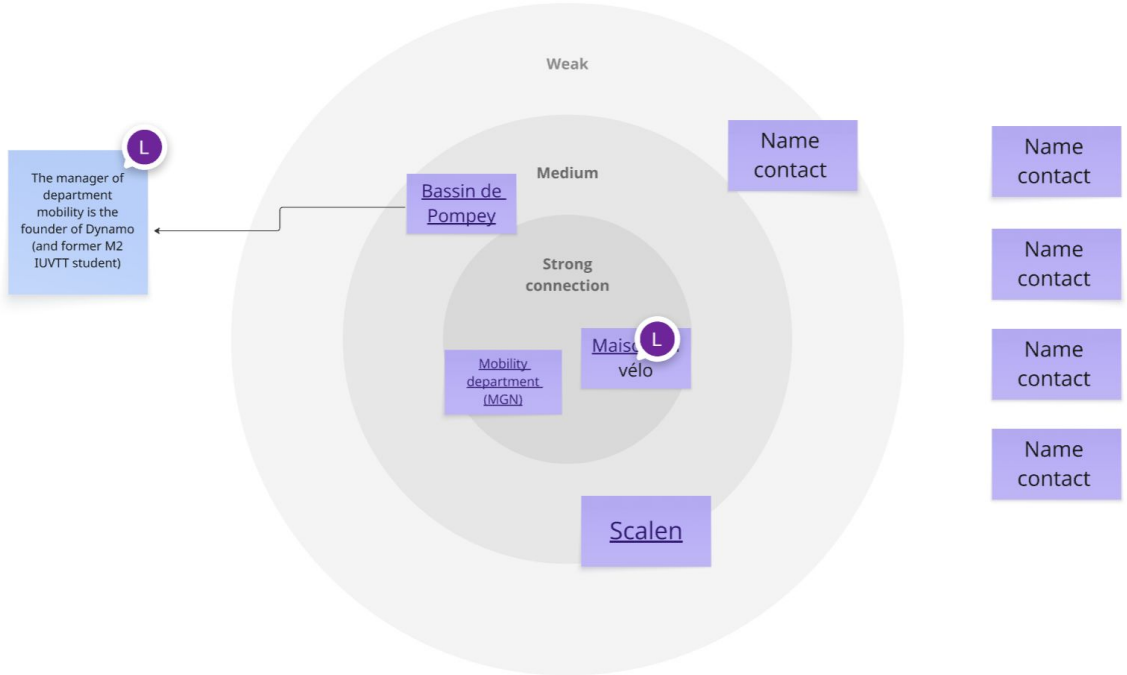
Name contact

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Name contact

Politicians, Municipal staff/departments, etc



Business



Name contact

Name contact

Name contact

Name contact

Name contact

Name contact

Validate hypotheses that can be dada-tested

Advocacy priorities for cycling in Lorraine

building of the advocacy campaign

H1 · Calm & cyclable neighbourhoods

Key ideas:

- Traffic calming (one-way, 30 km/h zones, modal filters).
- Protected cycling infrastructure.
- More vegetation and improved public spaces.
- Strong demand for transparent, participatory planning processes.

Data we plan to explore:

- Quality of surfaces **
- Close-passing incidents (“taking over manoeuvres”).**
- Pollution hotspots **
- % of routes without protected infrastructure.*

H2 · Street space towards bikes and active modes

Key ideas:

- More space for bikes and pedestrians
- Creation of continuous cycling corridors
- Better connections between centre and peri-urban areas.
- Stronger integration between cycling and public transport systems.

Data we plan to explore:

- Bicycle infrastructure *
- Car space x Bike space (área) *
- Hotspot of cyclists standing **
- Cyclist speed

* = (confirm city dataset)
** = senseBox:bike

Advocacy priorities for cycling in Lorraine

building of the advocacy campaign

H3 • Night-time safety and visibility as key determinants of cycling adoption

Key ideas:

- Lighting gaps on major cycling routes (canal towpaths, underpasses, river paths).
- Cyclists' perception of insecurity at night.
- Potential for visibility support stations (lights, reflectors, small repair tools).

Data we plan to explore:

- Mapping "dark spots" with light sensor;**
- Night-use patterns of main cycling routes;*
- City light infrastructure.*

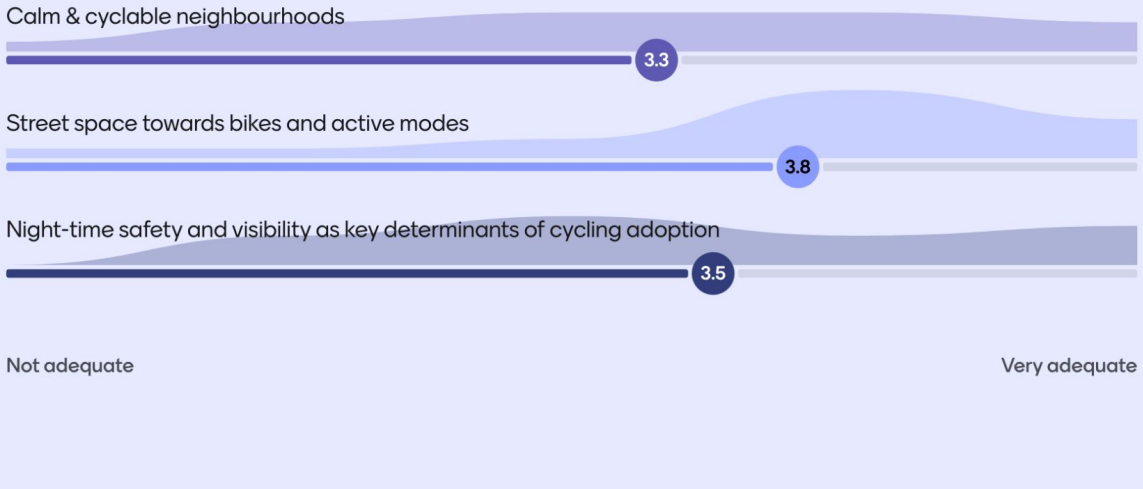
* = (confirm city dataset)
** = senseBox:bike

You have results from **15 participants.**

View results Clear results

Mentimeter

Hypotheses collective prioritisation



Installation of the prototype in the bikes and data collection

+

Community Sensing Campaign with local cyclists

Device installation







Initial discussion with mapped stakeholders + Focus Group



Emerging Findings

The adoption and everyday use of cycling depend less on the existence of isolated bicycle lanes and more on the continuity, width, and design quality of a coherent cycling network.

Participants repeatedly emphasized that “**good enough infrastructure**”, if continuous and predictable, may be more important than fragmented high-quality segments.

DATA-RIDE can explore how these spatial characteristics relate to:

- Overtaking distances between cars and bicycles
- Available corridors for the cyclists
- Speed variations and interruptions along routes
- Hotpots of standing
- Path conditions - roughness

Updated Hypothesis

H1 · The continuity and design quality of cycling infrastructure are decisive to adopting bikes as a means of transport

The adoption and everyday use of cycling depend less on the existence of isolated bicycle lanes and more on the continuity, width, and design quality of a coherent cycling network.

Discussions with cyclists and local stakeholders highlighted several issues affecting perceived safety and usability:

- discontinuous cycling lanes
- inconsistent lane markings
- narrow infrastructures squeezed between parked cars and traffic
- bicycle lanes placed on streets where spatial constraints create uncomfortable or unsafe riding conditions

Updated Hypothesis

H2 · Cycling safety emerges from interactions between mobility modes

Cycling safety is shaped not only by infrastructure but also by dynamic interactions between cyclists, cars, pedestrians, and micro-mobility users.

Focus group discussions highlighted multiple sources of perceived danger:

- proximity between cars and bicycles
- pedestrian behaviour in shared spaces (particularly near tram lanes)
- unpredictability of scooters and other micromobility devices
- signalling and awareness gaps among drivers (especially in crossroads)

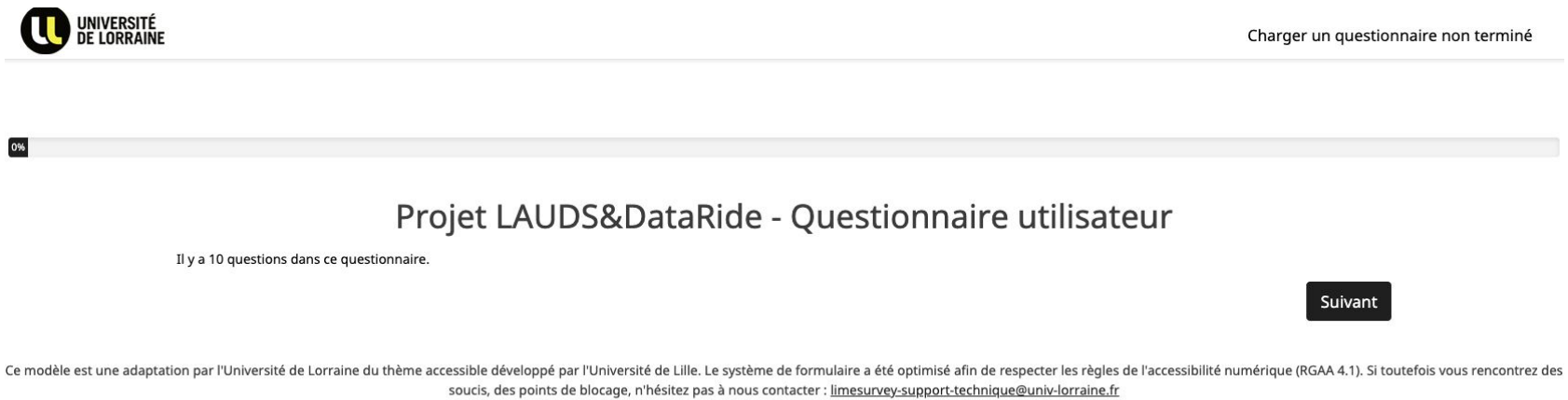
Campaign numbers

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Period: February 11 - April 30, 2026 (extended by 18.mai)

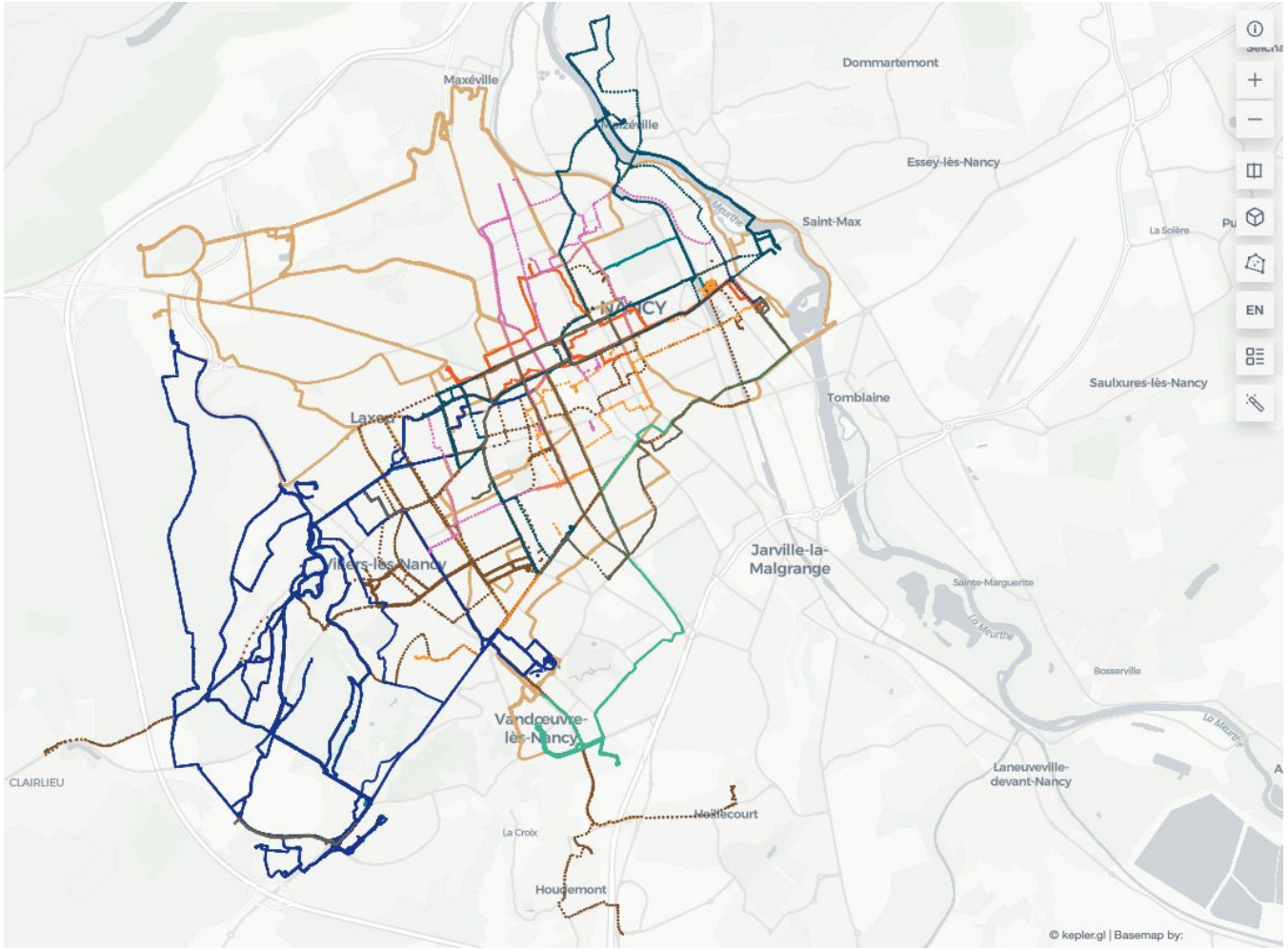
- 10 users / sensor boxes participated in the campaign
- 569 cycling trips recorded
- 1,906 km cycled in total
  (estimated from GPS movement, excluding a few unrealistic GPS jumps)
- 285,206 data points collected

Feedback : forms after the data collection



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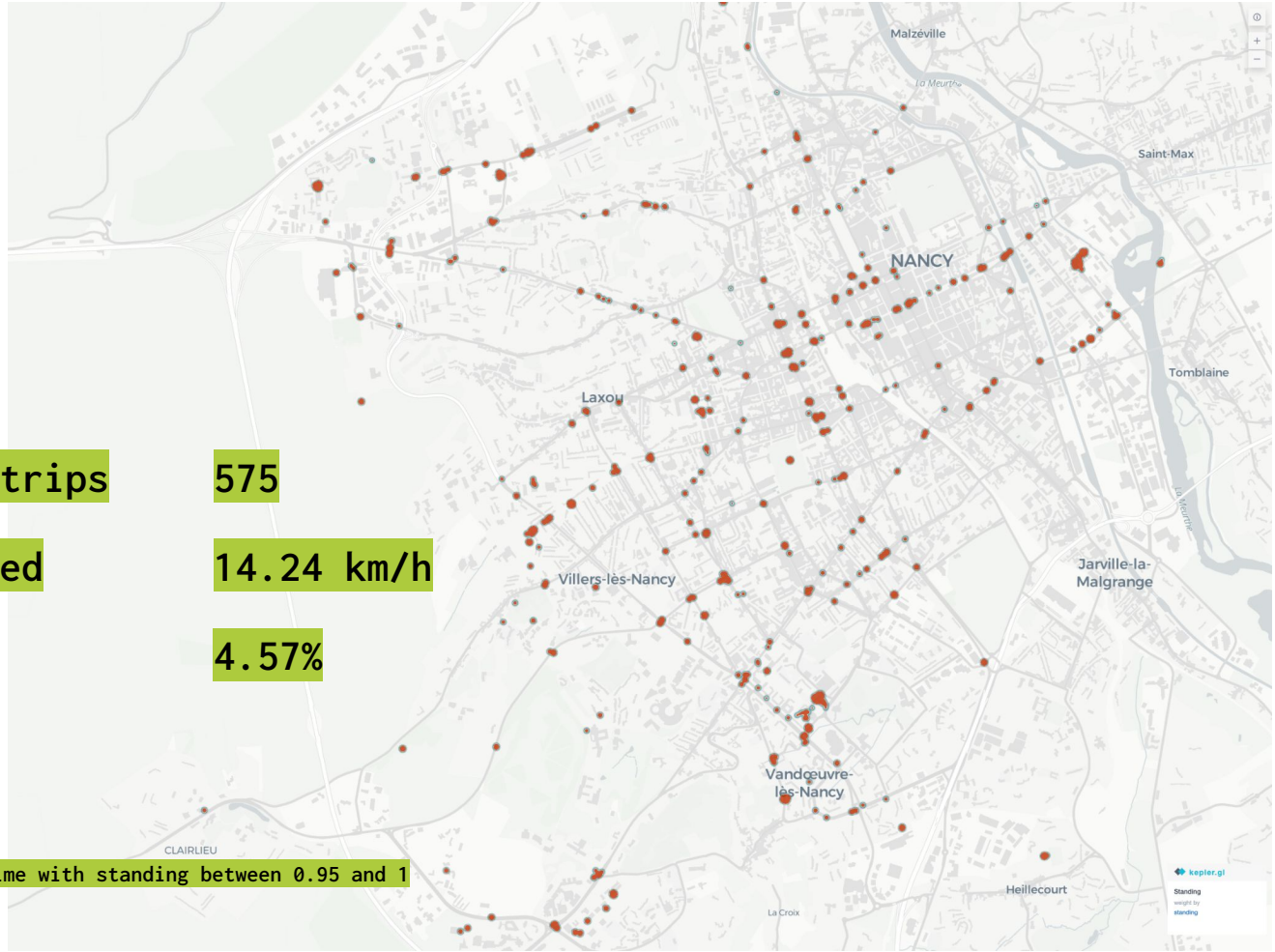
Data analysis : Rides per user



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Data analysis : Standing Hotspots

Total number of trips	575
Average trip speed	14.24 km/h
Standing*	4.57%



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*Share of time with standing between 0.95 and 1

Data analysis : bike path conditions



Smooth + Moderate
314.46 km* / 71.29%

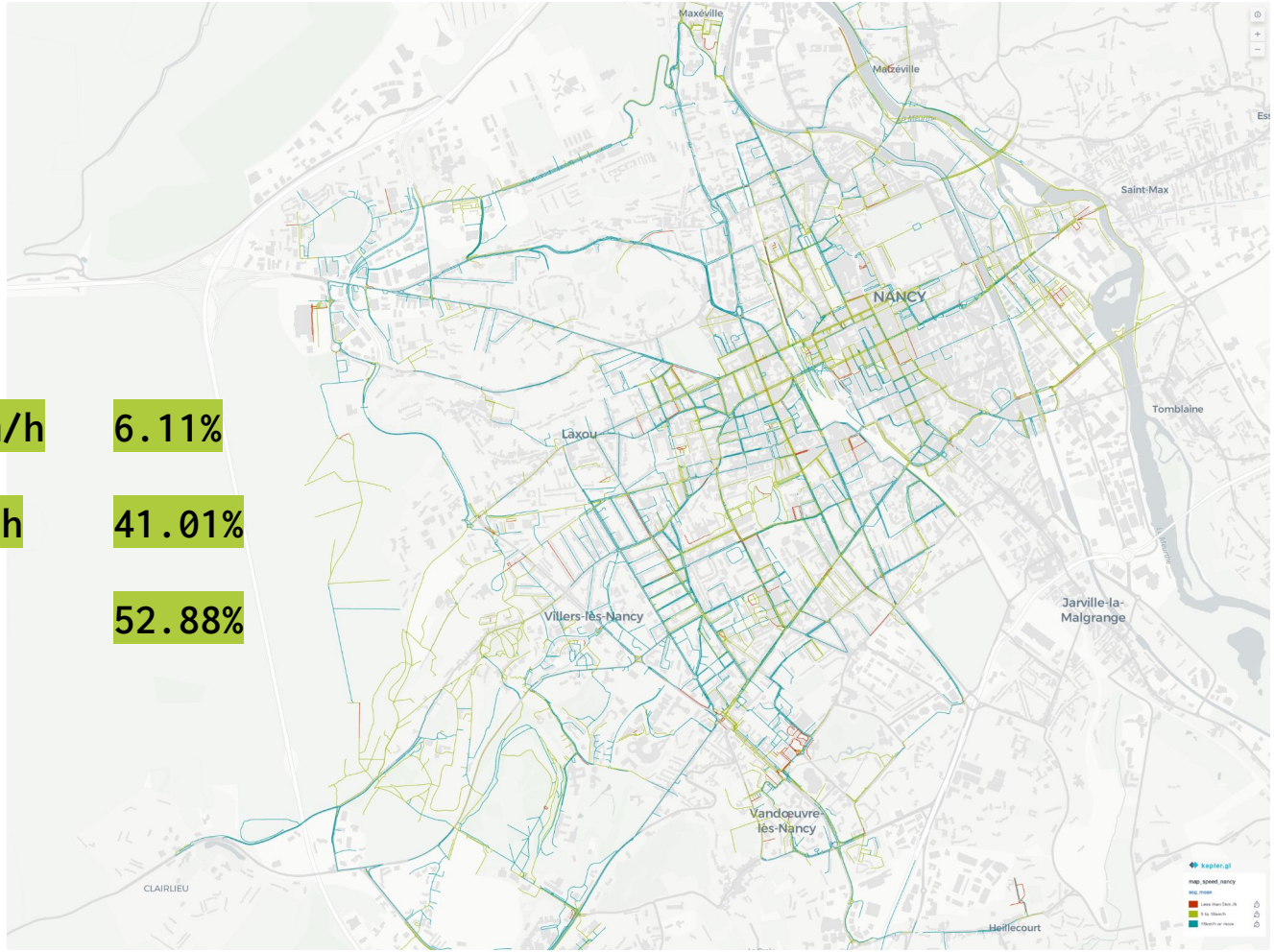
Rough + Very Rough
126.61 km* / 28.71%

*Percentages refer to the length of OSM road segments with sensor data, not the total distance cycled.

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Data analysis : Speed

very low speed < 5 km/h	6.11%
Reduced flow 5-15 km/h	41.01%
Fluid flow > 15 km/h	52.88%



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Data analysis : Corridors while taking over

Less than 1.50 m

90.4%

More than 1.50 m

9.6%



*Values refer to the minimum recorded overtaking distance, i.e. the closest passing distance measured for each event

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Data analysis : Overtaking events

54 likely true close-passing events below 1.50 m per 100 km cycled



Across an estimated 1,467.71 km cycled by the sensor-equipped bicycles, approximately 787 likely true overtaking events below 1.50 m were estimated after applying the Porto precision benchmark. This corresponds to roughly 54 likely close-passing events per 100 km cycled.

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Data analysis : Overtaking distance



Less than 1.50 m

96.77%

More than 1.50 m

3.23%

*Values refer to the minimum recorded overtaking distance, i.e. the closest passing distance measured for each event

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Our perspective

We propose to collectively discuss and further develop Hypothesis #2

H2 · Cycling safety emerges from interactions between mobility modes

Cycling safety is shaped not only by infrastructure itself, but also by the dynamic interactions between cyclists, cars, pedestrians, and micro-mobility users.

Findings

- 90.4% of corridor overtaking situations occur with less than 1.50 m distance
- 96.77% of overtaking events happen below the recommended 1.50 m safety distance

Implication

These findings support the legitimacy of advocating for 1.5 m+ overtaking distance and safer spatial separation between cars and cyclists as a relevant public safety measure.

Next step

This can serve as the starting point for the DATA-RIDE Workshop in Nancy (01.06.2026), where these findings and hypotheses can be collectively discussed, refined, and transformed into the basis for an advocacy campaign

Data processing: Methodology

1. Data Acquisition

Download CSVs from openSenseMap
One file per device
Time-filtered data

2. Standardization & Cleaning

Merge all files → single dataset
Create trips (gap > 5 min)
Remove:
invalid GPS
first/last 15s
distances <30 cm or >300 cm
Compute:
corridor width
roughness index (0-100)

3. Event Detection

Candidate detection (3s window):

- dmin ≤ 250 cm
- 30 ≤ variation ≤ 120 cm

Cleaning:

- remove stationary (standing = 1)
- remove parked-car patterns

Final events:

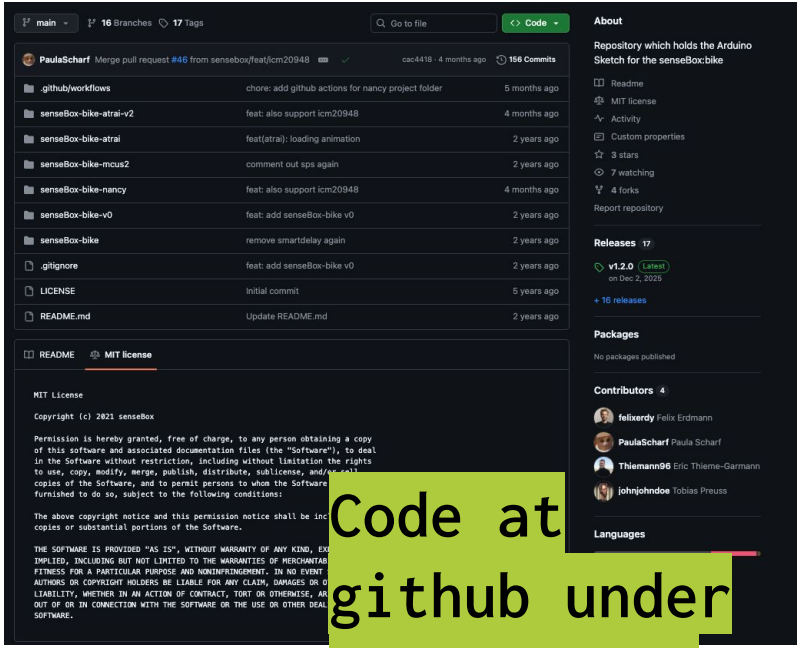
- merge ±1s
- keep closest interaction

Output

- Overtaking events (medium + high confidence)
- Ready for QGIS & spatial analysis

DATA RIDE

Hardware and open source information



Code at github under MIT License



PDF with the assembly manual

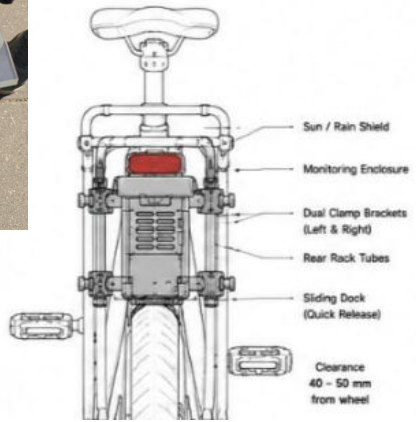
Hardware and open source information

box_id	trip_id	timestamp_qgis	latitude	longitude	trip_direction	roughness_index	distance_left	overtaking_manoeuvre	distance_right	finedust_pm1	finedust_pm10	finedust_pm2_5	finedust_pm4	speed	standing
box_01	1	2026-02-11 18:53:48	48.6940066	6.1996575	outbound	5.25	34.0	0.89	53.0	0.5	1.01	0.74	0.93	0.0	0.85
box_01	1	2026-02-11 18:53:54	48.6939951	6.1996471	outbound	1.0	31.0	0.98	34.0	0.72	1.43	1.06	1.31	0.0	0.96
box_01	1	2026-02-11 18:54:08	48.6939215	6.199503	outbound	33.5	64.0	0.09	32.0	0.76	1.5	1.11	1.38	0.65	0.03
box_01	1	2026-02-11 18:54:11	48.6939459	6.1995565	outbound	1.25	73.0	0.0	107.0	1.19	2.35	1.74	2.16	0.18	0.95
box_01	1	2026-02-11 18:54:13	48.6939847	6.1995826	outbound	0.0	76.0	0.68	107.0					0.0	
box_01	1	2026-02-11 18:54:14	48.6939787	6.1995982	outbound	0.75	77.0	0.76	104.0	1.66	3.25	2.42	2.99	0.0	0.98
box_01	1	2026-02-11 18:54:15	48.6939231	6.1995338	outbound	0.0	68.0	0.0	114.0					0.0	
box_01	1	2026-02-11 18:54:18	48.6939551	6.1995659	outbound	3.5	72.0	0.0	97.0	1.76	3.45	2.56	3.16	0.0	0.9
box_01	1	2026-02-11 18:54:21	48.6939612	6.1995752	outbound	0.0	68.0	0.54	91.0					0.0	
box_01	1	2026-02-11 18:54:22	48.6939647	6.1995761	outbound	0.0	74.0	0.54	116.0					0.0	
box_01	1	2026-02-11 18:54:23	48.693966	6.1995749	outbound	5.0	86.0	0.01	113.0	1.95	3.79	2.83	3.49	0.0	0.83
box_01	1	2026-02-11 18:54:24	48.6939734	6.1995538	outbound	0.0	115.0	0.03	110.0					1.02	
box_01	1	2026-02-11 18:54:26	48.6939855	6.1995163	outbound	5.75	47.0	0.01	120.0	2.13	4.11	3.07			
box_01	1	2026-02-11 18:54:28	48.6939731	6.1995251	outbound	0.0	35.0	0.01	140.0						
box_01	1	2026-02-11 18:54:31	48.6939707	6.19952	outbound	0.0	122.0	0.01	156.0						
box_01	1	2026-02-11 18:54:34	48.6939709	6.1995154	outbound	0.0	35.0	0.02	130.0						
box_01	1	2026-02-11 18:54:37	48.6939884	6.1995246	outbound	0.0	151.0	0.02	135.0						
box_01	1	2026-02-11 18:54:41	48.6939943	6.199523	outbound	0.0	37.0	0.02	125.0						
box_01	1	2026-02-11 18:54:43	48.6939932	6.1995157	outbound	0.25	112.0	0.02	129.0	2.62	4.93	3.7		0.28	0.99

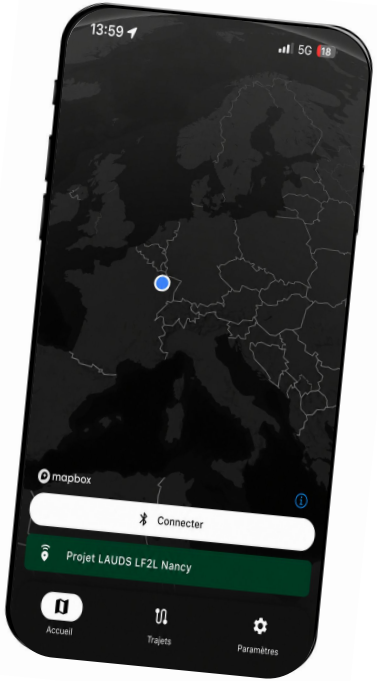
Anonymized users - boxID

Mounting system improvements - 3D printing





Software update in French



Main insights

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- **10 participants took part in the pilot phase.**
- **5 unique boxes were used during the test period in two rounds**
- **The seat-post mount was the dominant setup.**
- **The most frequent issues were:**
  - Overtaking detection reliability
  - Connectivity problems
  - Device size and portability
  - Mounting system limitations
  - Visibility and theft concerns

- **Positive feedback highlighted:**
  - Good battery performance
  - Easy-to-use app
  - Manageable data upload
  - Interest in testing an improved version

Next steps

- Mounting prototype - test the two versions during the Trip to Nancy
- Discuss, organize and publish the materials (dataset, data processing methodology, maps, advocacy materials, policy recommendation) on the Lauds Website
- Advocacy campaign kick-off and preparation of a draft for policy recommendations - workshop in Nancy
- Develop the tool kit (format and final template)

Next steps

Agenda - Travel to Nancy: 28 May - 3 June

30/05 - Participation in Fête du Vélo
Dissemination of the project and initial engagement with local actors and the wider public.

01/06 - Presentation + Workshop
Presentation of collected data and insights, followed by a collaborative workshop to analyse findings and initiate the co-creation of the advocacy campaign with motivated local actors.

02/06 - Collaborative Work Session
Further development of the advocacy campaign strategy, including:

- Communication materials (posters, stickers, online dissemination)
- Outreach strategy (including potential engagement with policymakers)

Wrap-up Session
Collective reflection on the sustainability of the advocacy process, including:

- Consolidation of key insights
- Planning the next steps
- Preparation of the final policy recommendations document

Merci
Danke
Thanks
Obrigada

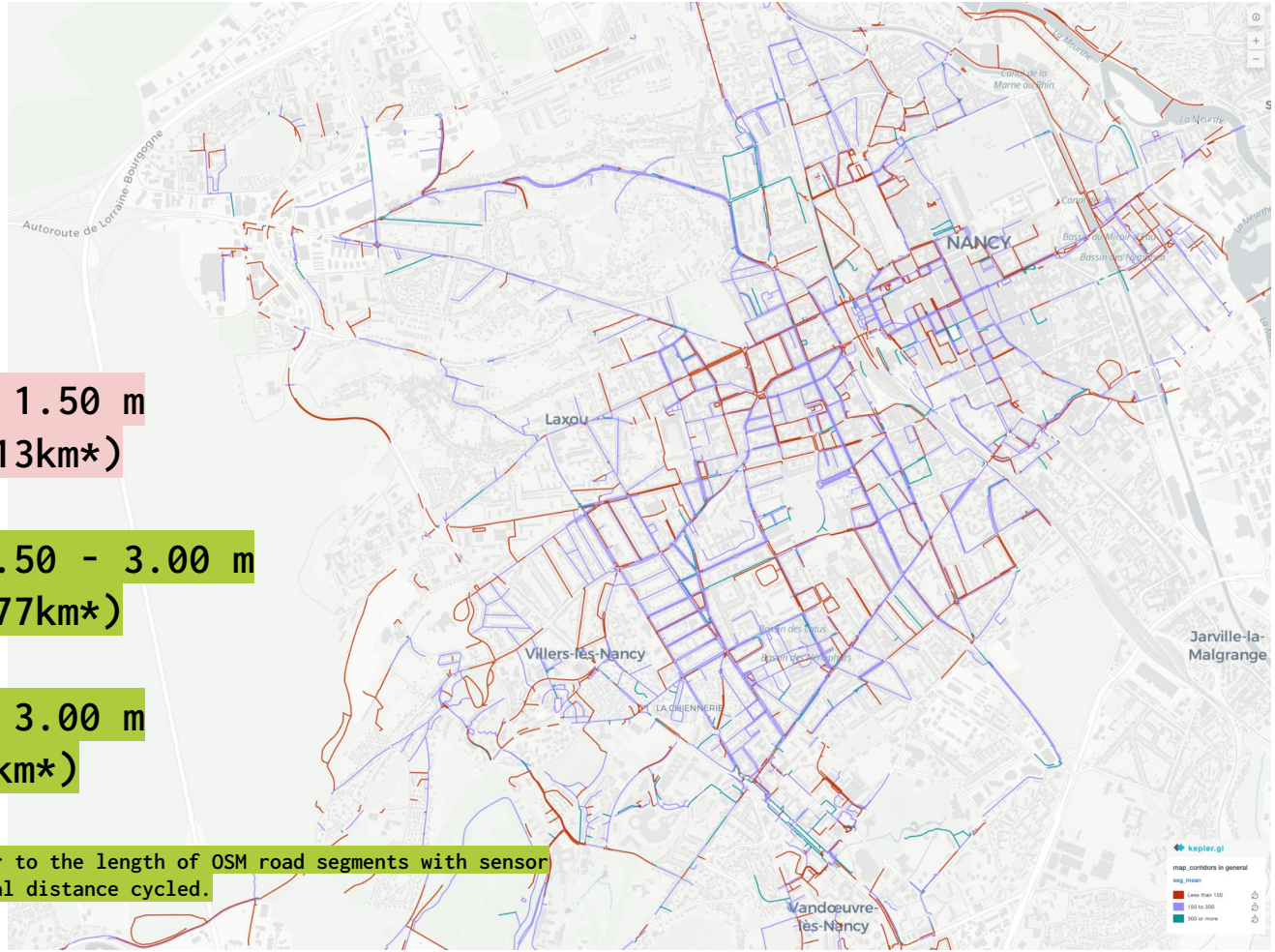


Data analysis : corridors

Less than 1.50 m
52.37% (213km*)

Between 1.50 - 3.00 m
43.73% (177km*)

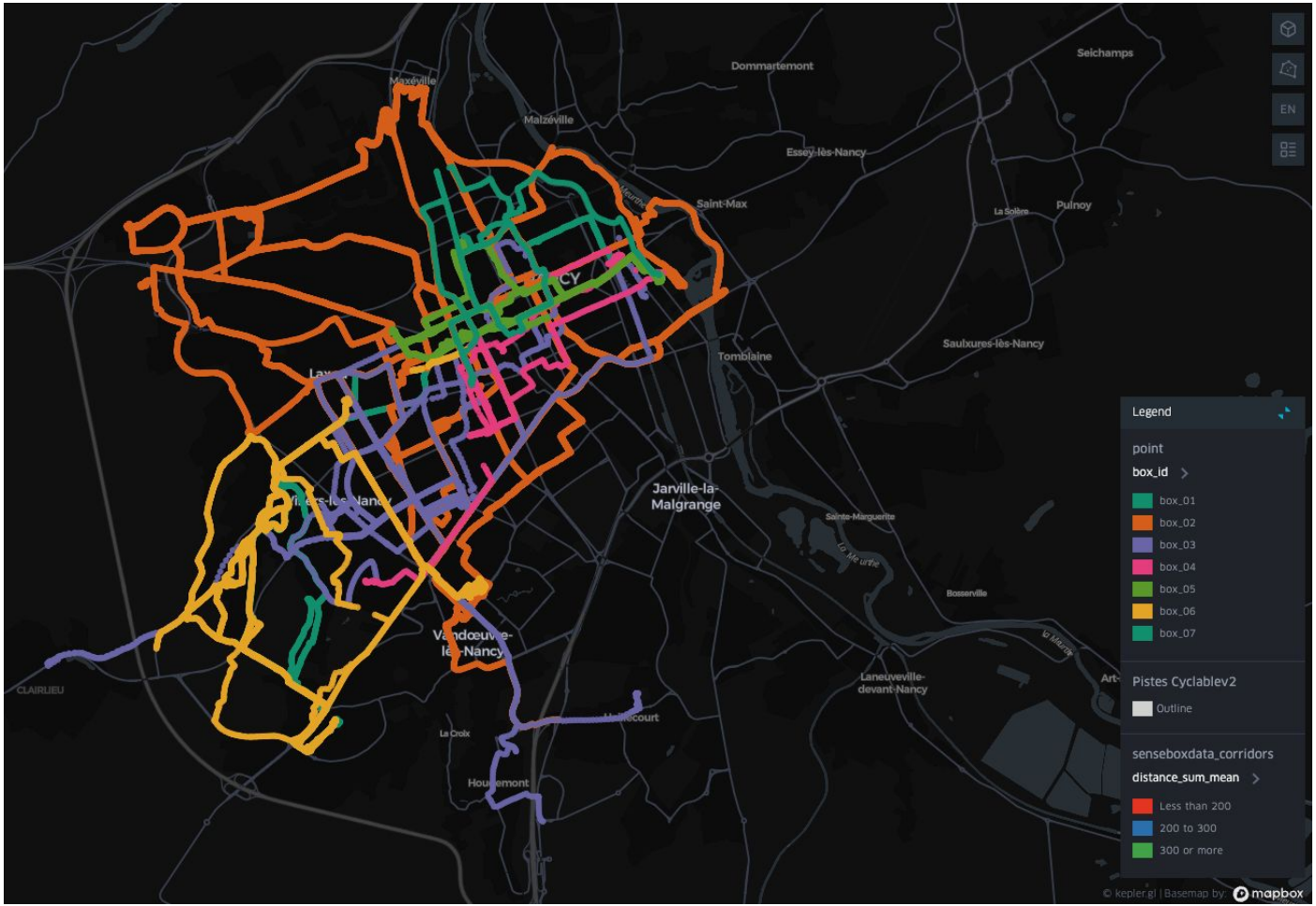
More than 3.00 m
3.90% (15km*)



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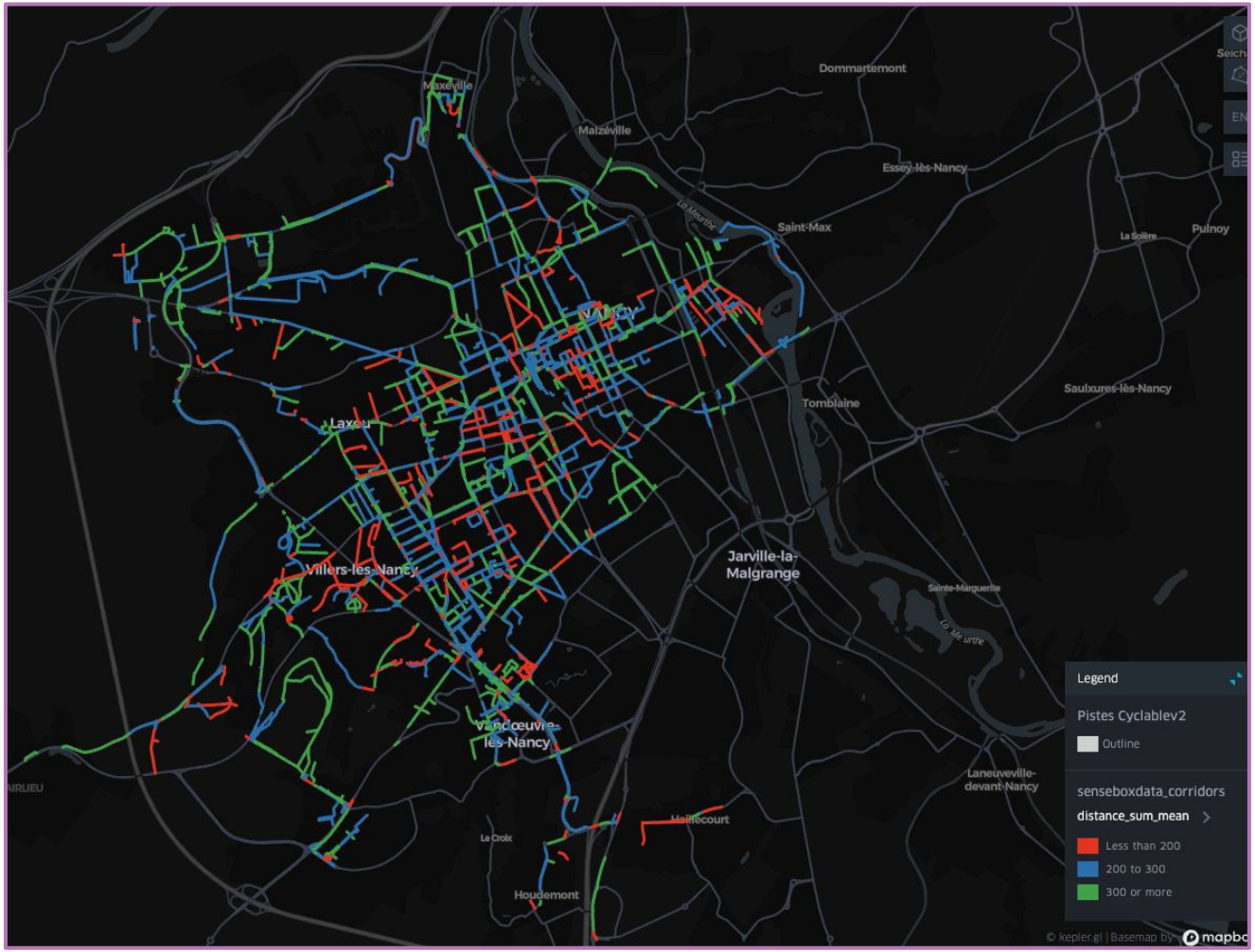
*Percentages refer to the length of OSM road segments with sensor data, not the total distance cycled.

Data analysis : Rides per user



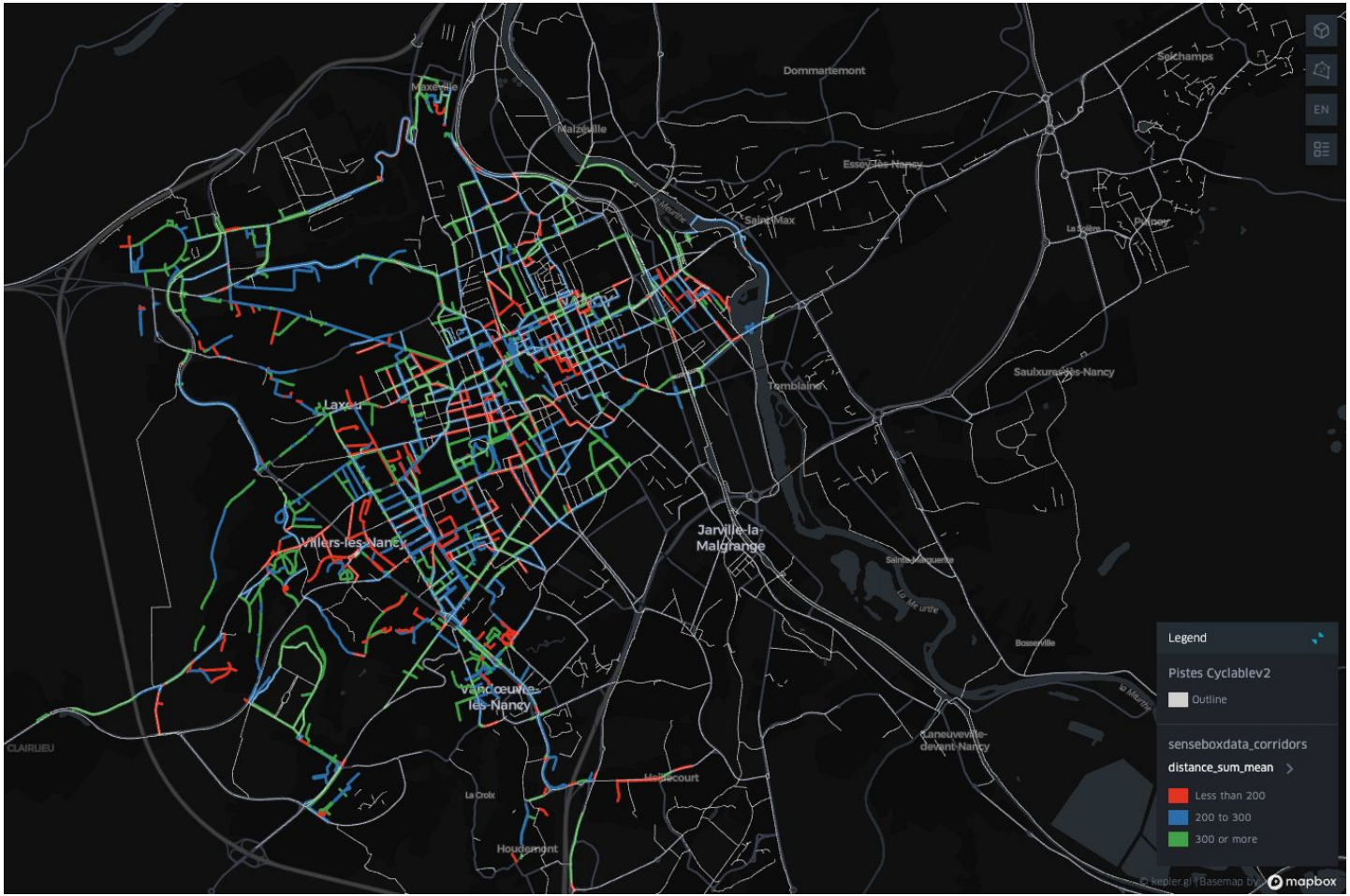
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Data analysis : Corridors



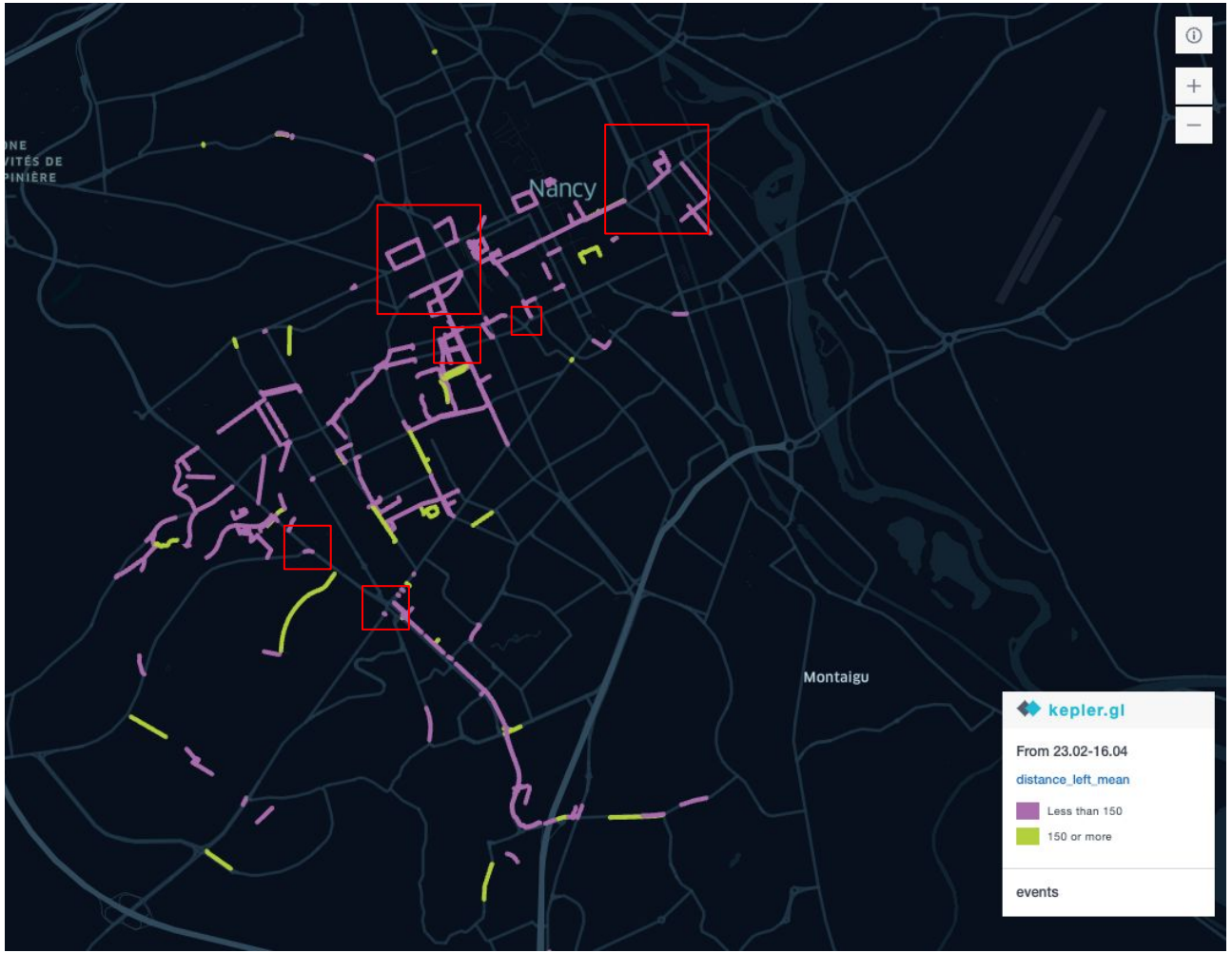
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Data analysis : Corridors + Pistes Cyclable



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Data analysis : Taking over distance



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Data analysis : bike path conditions

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Participant	Smartphone	Box ID(s)	Mounting	Main feedback
Benoît Pouyatos	iPhone 14	3 boxes	Seat post	Several issues reported
Hugo Pierson	Google Pixel 9A	1 box	Seat post	Some disconnections; overtaking detection needs improvement
Alex Gabriel	Samsung	1 box	Seat post	One reconnection needed
Luc Jansen	iPhone	1 box	Seat post	Repeated disconnections due to vibrations
Jules Bagot	iPhone	1 box	Seat post	Device perceived as bulky, but battery and mounting worked well
Marie Blanchard	Fairphone	1 box	Rear rack strap	Connection issues; mounting system difficult to use

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Indicator	Value
Participants involved	6
Total box assignments	8
Unique box IDs used	7
Seat-post mounting	5 participants
Rear-rack mounting	1 participant
Confirmed box swaps	2
Confirmed reconnection manipulations	3
Participants reporting issues	5 / 6